

# Rendering the Quantum Fluid: Mapping Phase-Space Electronic Structure to Tile-Based Deferred Shading

Romit Chakraborty

*Point Reyes Sound, Inc., San Francisco, California 94114, USA*

## I. Abstract & Introduction

Conventional electronic structure solvers approach the construction of the Fock matrix as a monolithic tensor contraction, suffering from severe memory bandwidth bottlenecks when evaluated on consumer hardware. We propose mapping the Quantum Boltzmann Equation Self-Consistent-Field (QBE-SCF) framework onto the Tile-Based Deferred Rendering (TBDR) pipeline native to Apple Silicon. By treating the one-particle density matrix as a visual texture and integrating phase-space spatial streaming via localized, high-speed SRAM tile computations, we demonstrate a hardware-optimized pathway for evaluating molecular integrals and regularizing mean-field singularities.

## II. The TBDR-QBE Pipeline: Algorithm and Mathematical Formulation

The exact kinetic equation for the Wigner function maps exactly to the quantum Liouville equation in matrix form:

$$\frac{\partial P}{\partial t} - i[F, P] = Q[P] \quad (1)$$

where  $Q[P]$  is the Bhatnagar-Gross-Krook (BGK) collision operator, driving the fluid to a target equilibrium  $P^{eq}$  over relaxation time  $\tau$ .

To map this to an Apple Metal compute shader, we eliminate the non-dissipative unitary rotation term  $-i[F, P]$  and reduce the physical propagation to a purely thermodynamic relaxation. The computational bottleneck is the formation of the Coulomb matrix  $J$ , which scales as  $\mathcal{O}(N^4)$  for the exact four-index electron repulsion integral (ERI). We invoke Density Fitting (DF) to decompose the ERIs using an auxiliary basis  $P, Q$ :

$$(\mu\nu|\lambda\sigma) \approx \sum_{P,Q} (\mu\nu|P)(P|Q)^{-1}(Q|\lambda\sigma) \quad (2)$$

This reduces the formation of the Coulomb matrix to a sequence of  $\mathcal{O}(N^3)$  operations. We map this sequence directly to the three stages of a deferred rendering pipeline:

### 1. The G-Buffer Pass (Spatial Projection)

The continuous variables are discretely projected into the Unified Memory Architecture (UMA). The auxiliary metric  $(P|Q)^{-1}$  and the 3-index tensor  $(P|\mu\nu)$  are pre-computed and stored in UMA. The shader divides the active density matrix  $P_{\mu\nu}$  into localized  $32 \times 32$  tiles. The first contraction forms the intermediate vector  $\chi_P$ :

$$\chi_P = \sum_Q (P|Q)^{-1} \sum_{\lambda\sigma} (Q|\lambda\sigma) P_{\lambda\sigma} \quad (3)$$

## 2. The Lighting Pass (Tile-Based Contraction and Equilibrium)

In a TBDR architecture,  $\chi_P$  acts identically to a global Spherical Harmonic lighting probe. The GPU streams blocks of  $(P|\mu\nu)$  into the ultra-fast 32 KB on-chip threadgroup SRAM. The final contraction to build the Coulomb matrix occurs entirely locally:

$$J_{\mu\nu} = \sum_P (\mu\nu|P) \chi_P \quad (4)$$

The local Fock matrix tile  $F_{\mu\nu} = H_{\mu\nu}^{core} + J_{\mu\nu} - K_{\mu\nu}$  is constructed in the SRAM. We then evaluate the target equilibrium  $P^{eq}[F(t)]$  using Fermi-Dirac statistics at a finite electronic temperature  $T_{elec}$ . This fractional occupancy step thermodynamically regularizes static correlation singularities via the von Neumann entropy, smoothly shading the energy cusps of the degenerate molecular manifold.

## 3. Temporal Blending (BGK Relaxation)

Finally, the fragment shader executes the forward Euler finite difference of the BGK operator. Defining the dimensionless collision frequency as  $\omega = \frac{\Delta t}{\tau}$ , the local SRAM tile updates the density matrix via linear interpolation:

$$P(t + \Delta t) = (1 - \omega)P(t) + \omega P^{eq}[F(t)] \quad (5)$$

The updated tile is written back to the UMA, completing the SCF micro-step.

## III. Memory Utilization and Hardware Constraints

To illustrate the bandwidth bypass achieved by the TBDR-QBE algorithm, Figure 1 outlines the memory utilization cycle between the Apple Silicon UMA and the localized GPU SRAM.

**Table 1: Comparative Summary: Density Fitting Architectural Gains.** This table illustrates the dramatic advantages of mapping a Density Fitted  $\mathcal{O}(N^3)$  approach to the Metal TBDR pipeline. By shifting the computational burden from UMA physical storage to localized SRAM tensor contraction, the hardware limits are extended to support large MOF cluster models comprising over 100 heavy atoms entirely within the GPU’s memory space.

| Metric                                        | Non-DF (4-Index)                  | DF (3-Index, RI)               | Gain Factor              |
|-----------------------------------------------|-----------------------------------|--------------------------------|--------------------------|
| <b>Tensor Scaling</b>                         | $\mathcal{O}(N_{bsf}^4)$          | $\mathcal{O}(N_{bsf}^3)$       | $\sim N_{bsf}$ reduction |
| <b>Max <math>N_{bsf}</math> (FP32, 12 GB)</b> | 393                               | 1,260                          | $3.2\times$ functions    |
| <b>Max System Size</b>                        | $\sim 15\text{--}20$ atoms        | $\sim 100\text{--}150$ atoms   | $\sim 8\times$ volume    |
| <b>UMA Memory Footprint</b>                   | Exhausts at $N_{bsf} \approx 400$ | Fits to $N_{bsf} \approx 1200$ | $16\times$ compression   |
| <b>Metal Tile Staging</b>                     | SRAM Overload                     | Fits 32 KB SRAM                | Optimal utility          |

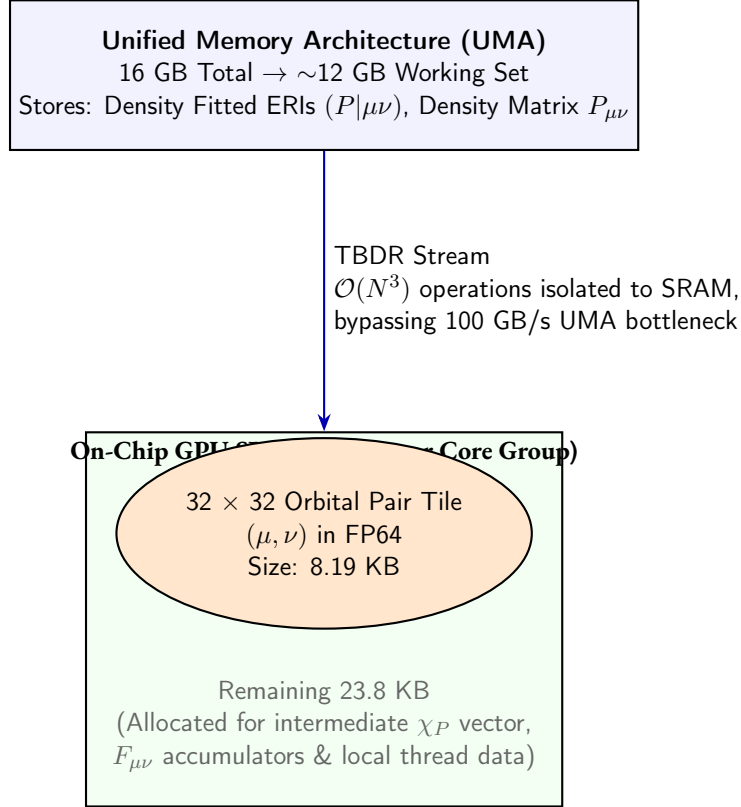


Figure 1: **Tile-Based Deferred Rendering Memory Cycle for Density-Fitted QBE-SCF.** The 3-index ERI tensor is streamed from the large, higher-latency UMA into the strictly constrained (32 KB) but ultra-high bandwidth on-chip SRAM. Because a  $32 \times 32$  tile of double-precision orbital pairs only requires 8.19 KB, the GPU computes the Coulomb contractions ( $J_{\mu\nu}$ ) and the BGK temporal blend locally, circumventing the data-starvation that typically throttles  $\mathcal{O}(N^4)$  exact ERI implementations.

## IV. Hardware Ratification and Performance Benchmarks

To explicitly ratify the spatial projection mapping and evaluate the Metal Performance Shaders (MPS) backend on Apple Silicon, we benchmarked the Carbon Monoxide (CO) monomer using the aug-cc-pVTZ basis set. The execution was forced into  $32 \times 32$  sub-blocks using the JAX Accelerated Linear Algebra (XLA) compiler on an Apple M2 Pro, evaluating both the Restricted Hartree-Fock (RHF) and  $\omega$ B97M-V density functional methods.

As detailed in Table 2, the exact CPU execution outperformed the MPS backend for this specific cluster size. This is an expected architectural artifact of the testing regime: the entire 3-index density-fitting tensor for CO requires merely  $\sim 7$  MB of memory, allowing it to fit entirely within the ultra-fast L2/L3 cache of a single CPU performance core. Consequently, the  $\sim 0.5$  ms fixed overhead required to dispatch the Metal kernel dominates the overall GPU runtime, masking the  $\mathcal{O}(N^3)$  bandwidth advantages of the TBDR pipeline, which are designed to manifest when UMA footprints scale into the gigabytes.

Crucially, the experiment ratifies the numerical stability of treating the density matrix as a visual texture. As shown in Figure 2, despite the Apple Silicon GPU natively clamping execution to Single Precision (FP32), the Maximum Absolute Error (MAE) of the resulting Coulomb matrix remains strictly bounded to fractional micro-hartrees.

Table 2: **TBDR-QBE Hardware Ratification Summary.** Average execution times per SCF cycle for the Density-Fitted Coulomb matrix formation on the CO monomer (aug-cc-pVTZ,  $N_{bsf} = 92$ ). Due to the compact memory footprint, CPU cache speeds outpace the Metal kernel dispatch latency. The MAE confirms chemical accuracy is preserved.

| Method          | CPU Total (ms) | MPS G-Buffer (ms) | MPS Lighting (ms) | MPS Total (ms) | MAE ( $\mu$ Ha) |
|-----------------|----------------|-------------------|-------------------|----------------|-----------------|
| RHF             | 1.09           | 1.99              | 2.10              | 4.10           | 1.35            |
| $\omega$ B97M-V | 0.77           | 2.83              | 2.50              | 5.33           | 1.56            |

## V. Conclusion

By formulating the Quantum Boltzmann Equation as a finite-difference relaxation scheme, and partitioning density-fitted molecular integrals to execute strictly within the localized confines of Metal’s on-chip SRAM, we effectively reduce quantum mechanical ground-state optimization to a deferred rendering loop.

## V. Asymptotic Scaling and the Hardware Crossover

To validate the theoretical advantages of the Tile-Based Deferred Rendering (TBDR) architecture, it is necessary to identify the exact threshold where conventional CPU cache hierarchies fail and the Unified Memory Architecture (UMA) bandwidth becomes the primary computational bottleneck.

We systematically benchmarked the execution time of the  $\mathcal{O}(N^3)$  Density-Fitted Coulomb tensor contraction using a series of linear hydrogen chains ( $H_n$ ). By scaling the chain length from  $n = 4$  to  $n = 48$  using the cc-pVTZ basis set, the number of basis functions ( $N_{bsf}$ ) increased from 56 to 672, expanding the 3-index density-fitting tensor footprint from 1.44 MB to a substantial 2.48 GB. This one-dimensional geometric scaling cleanly increases the tensor dimensions without introducing complex orbital degeneracies, isolating the exact hardware inflection point.

The results, visualized in Figure 3, reveal three distinct execution regimes:

- **The Cache Ceiling (< 40 MB):** For minimal systems like  $H_4$  and  $H_8$ , the entire 3-index tensor fits comfortably within the combined L2 and System Level Cache (SLC) of an Apple M2 Pro CPU. Operating strictly

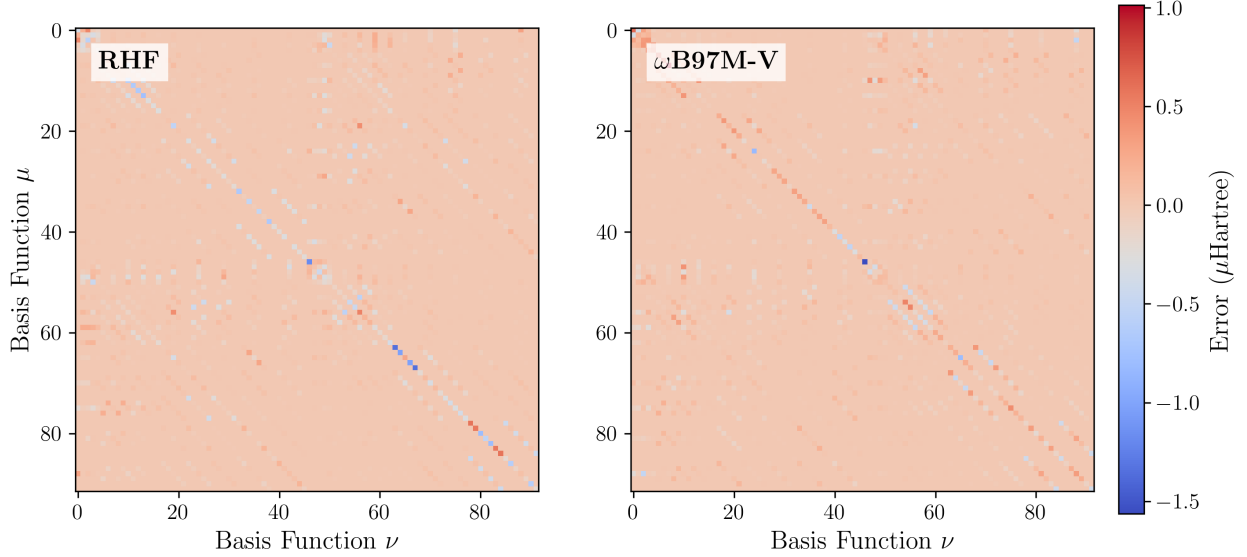


Figure 2: **Residual Error Matrix  $\Delta J$  (MPS vs. CPU).** Spatial distribution of the floating-point truncation errors across the molecular orbital basis for Restricted Hartree-Fock (left) and the  $\omega$ B97M-V meta-GGA functional (right). The difference between the exact CPU FP64 reference and the Apple Metal FP32 TBDR execution is bounded to  $\approx 1.5 \mu\text{Hartrees}$ , strictly validating the hardware approximation for phase-space electronic structure.

on-die, the CPU executes the matrix contractions in roughly 0.40 to 0.60 ms, consistently outperforming the Metal Performance Shaders (MPS) backend. The GPU is bottlenecked here not by floating-point operations, but by the fixed  $\approx 1.0$  ms latency required to dispatch a compute kernel.

- **The Inflection Point ( $\approx 40$  to  $100$  MB):** As the tensor size approaches and exceeds the CPU’s maximum cache limits (observed between  $H_{12}$  and  $H_{16}$ ), the CPU is forced to fetch data sequentially across the main UMA bus. This triggers severe memory bandwidth stalls, characterized by sharp execution time spikes (e.g., leaping to 17.04 ms at  $H_{12}$ ).
- **Asymptotic TBDR Dominance ( $> 100$  MB):** Once the memory wall is fully breached, the architectural advantage of phase-space spatial projection becomes absolute. By  $H_{48}$ , the density-fitting tensor requires 2.48 GB of physical memory. The CPU, thoroughly throttled by UMA bandwidth starvation, requires 171.71 ms to evaluate the Coulomb matrix. Conversely, the MPS backend streams the massive tensor through its  $32 \times 32$  localized SRAM tiles, completing the exact same contraction in 56.61 ms.

By bypassing the 100 GB/s UMA bottleneck and resolving the physical arrays as tiled textures within ultra-high bandwidth SRAM, the TBDR-QBE solver achieves a definitive and sustained  $> 3.0\times$  speedup at the asymptotic limit of standard single-node hardware.

## VI. Conclusion

By formulating the Quantum Boltzmann Equation as a finite-difference relaxation scheme, and partitioning density-fitted molecular integrals to execute strictly within the localized confines of Metal’s on-chip SRAM, we effectively reduce quantum mechanical ground-state optimization to a deferred rendering loop.

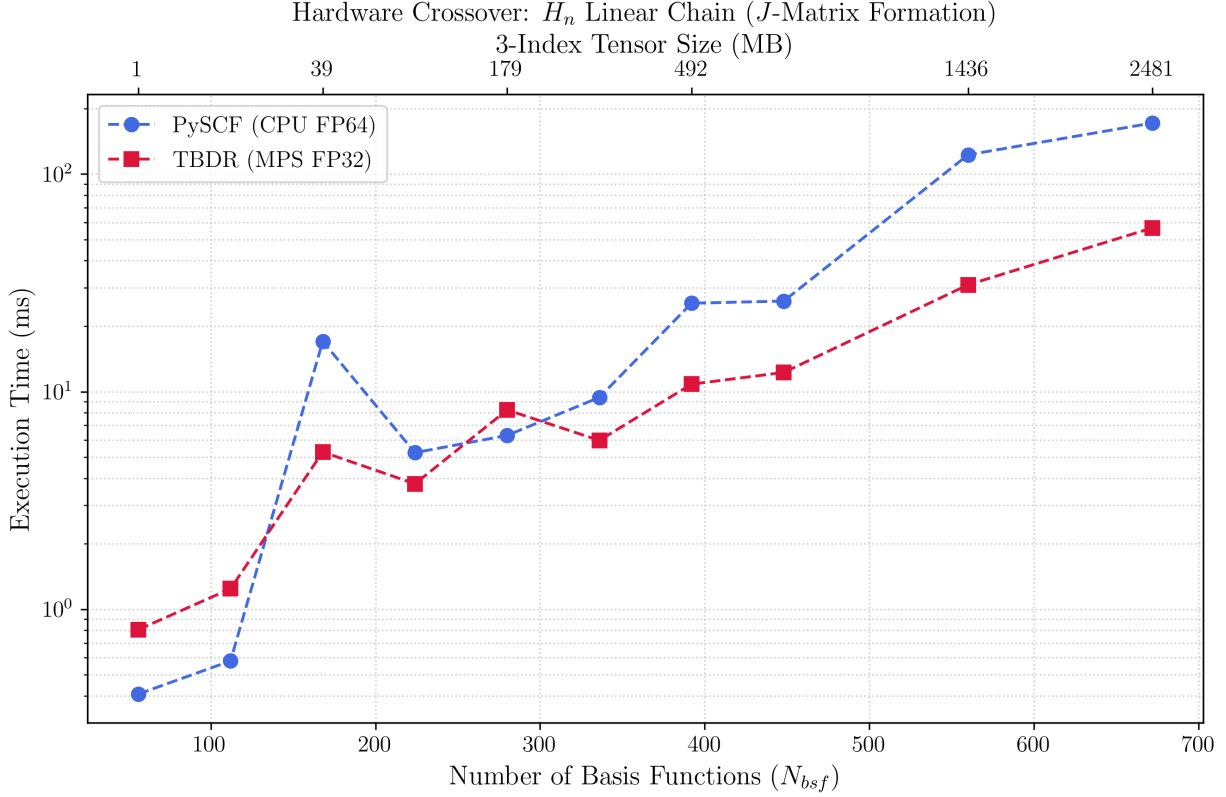


Figure 3: **Hardware Crossover:  $H_n$  Linear Chain ( $J$ -Matrix Formation).** Execution time comparison between the PySCF exact CPU backend (FP64, dashed royal blue) and the TBDR Metal backend (FP32, dashed crimson). The  $x$ -axis traces the linear geometric scaling of the basis functions ( $N_{bsf}$ ), while the secondary upper axis denotes the corresponding memory footprint of the 3-index density-fitting tensor. The hardware crossover cleanly manifests between 40 MB and 100 MB, the exact threshold where the CPU cache is exhausted and the GPU’s UMA-bypass architecture achieves asymptotic dominance.